
LECTURE NOTES

Lecture 14: Linear Constrained Optimization

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1 Linear Programming

1.1 Problem Formulation

If an optimization problem has a linear objective and constraints, it is called a **linear programming problem (linear program, LP)**

- The general form is

$$\begin{aligned} \min_x \quad & c^T x \\ \text{s.t.} \quad & Ax \leq b \\ & Dx \geq e \\ & Fx = G \\ & x, c \in \mathbb{R}^n, \\ & A \in \mathbb{R}^{m \times n}, b \in \mathbb{R}^m \\ & D \in \mathbb{R}^{p \times n}, e \in \mathbb{R}^p \\ & F \in \mathbb{R}^{q \times n}, g \in \mathbb{R}^q \end{aligned}$$

1.1.1 Numerical example

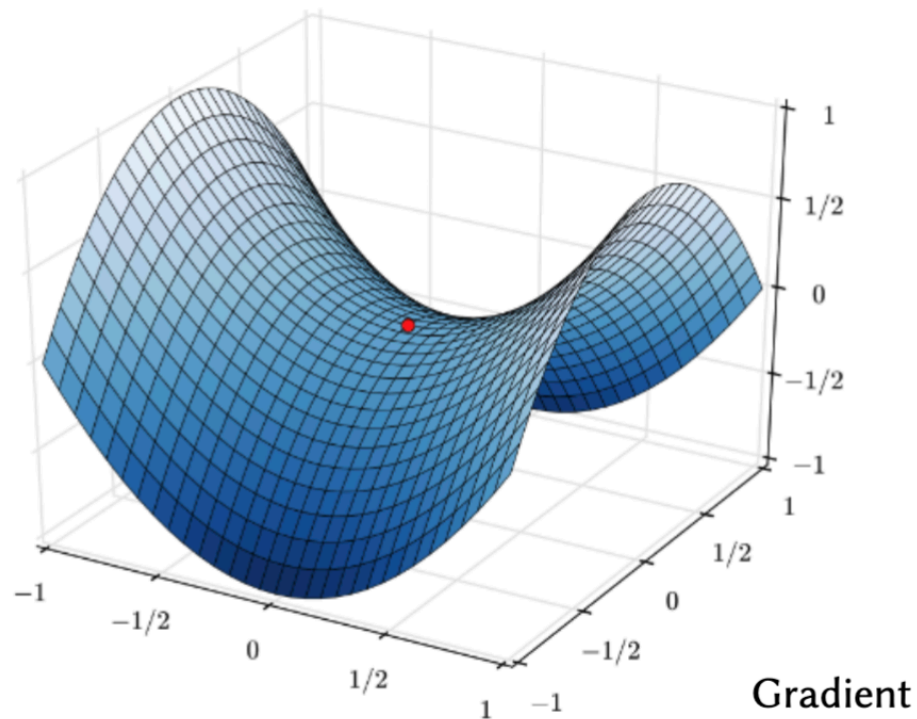
$$\begin{aligned} & \underset{x_1, x_2, x_3}{\text{minimize}} \quad 2x_1 - 3x_2 + 7x_3 \\ & \text{subject to} \quad 2x_1 + 3x_2 - 8x_3 \leq 5 \\ & \quad \quad \quad 4x_1 + x_2 + 3x_3 \leq 9 \\ & \quad \quad \quad x_1 - 5x_2 - 3x_3 \geq -4 \\ & \quad \quad \quad x_1 + x_2 + 2x_3 = 1 \end{aligned}$$

$$\begin{bmatrix} 2 & 3 & -8 \\ 4 & 1 & 3 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \leq \begin{bmatrix} 5 \\ 9 \end{bmatrix}$$

Where

$$x_1, x_2, x_3 \in \mathbb{R}$$

1.1.2 General example



Binary programming works better. So if you had to choose an amount of each item, you should make each item (if you have 10 phones = $\{x_1, \dots, x_{10}\}$) a parameter.

1.2 Conversion to linear program

Many problems can be converted into linear programs that have the same solution.

Example	Example
$\text{minimize } L_1 = \ A\mathbf{x} - \mathbf{b}\ _1$	$\text{minimize } L_\infty = \ A\mathbf{x} - \mathbf{b}\ _\infty$
$\begin{aligned} \min \mathbf{1}^T \mathbf{s} \\ \text{s.t. } A\mathbf{x} - \mathbf{b} \leq \mathbf{s} \\ \quad - (A\mathbf{x} - \mathbf{b}) \leq \mathbf{s} \end{aligned}$	$\begin{aligned} \min t \\ \text{s.t. } A\mathbf{x} - \mathbf{b} \leq t\mathbf{1} \\ \quad - (A\mathbf{x} - \mathbf{b}) \leq t\mathbf{1} \end{aligned}$
Sum of absolute errors Robust regression: Ignores outliers; focuses on the "median" fit.	Minimax Approximation or Chebyshev Approximation problem

1.3 Forms

Two canonical ways to write an LP. Start from the general form and convert to one of these to hand it to an algorithm.

1.3.1 Standard form

All constraints are inequalities, variables non-negative:

$$\begin{array}{ll}\min_x & c^T x \\ \text{s.t.} & Ax \leq b \\ & x \geq 0\end{array}$$

Conversion rules:

- $A \geq$ constraint is flipped by multiplying by -1
- An equality $Fx = g$ becomes two inequalities: $Fx \leq g$ and $-Fx \leq -g$
- A free variable $x_i \in \mathbb{R}$ is split: $x_i = x_i^+ - x_i^-$ with $x_i^+, x_i^- \geq 0$

Nice for geometric intuition — the feasible set is a polytope defined by half-spaces.

1.3.2 Equality form

Also called *standard equality form* or *slack form*. The **main constraints** are equalities; $x \geq 0$ is a separate sign restriction (non-negativity bound), not counted among the “constraints” in this naming:

$$\begin{array}{ll}\min_x & c^T x \\ \text{s.t.} & Ax = b \quad (\text{main constraints — equalities}) \\ & x \geq 0 \quad (\text{sign restriction})\end{array}$$

Conversion rules:

- $Ax \leq b$ becomes $Ax + s = b$ with slack $s \geq 0$
- $Ax \geq b$ becomes $Ax - s = b$ with surplus $s \geq 0$
- Free variables split the same way as above

This is the form the **simplex algorithm** operates on — it needs equalities so it can pick a basis A_B (an $m \times m$ invertible submatrix) and solve $x_B = (A_B)^{-1}b$ at each vertex. That is why the dimension grows to $2n + m$: n originals + n for splitting free vars + m slacks.

1.3.3 Quick example

Original problem:

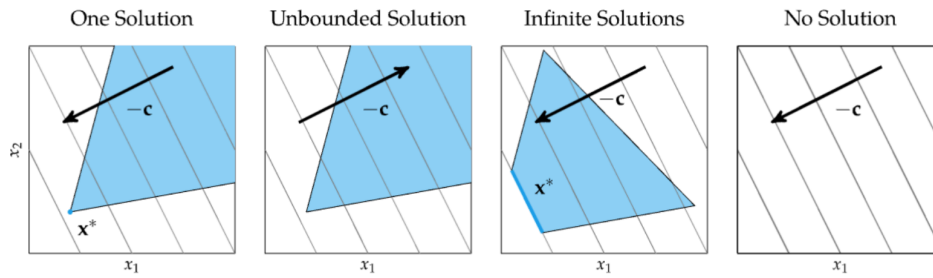
$$\min \quad x_1 + x_2 \quad \text{s.t.} \quad 2x_1 + x_2 \leq 4, \quad x_1, x_2 \geq 0$$

Equality form — add slack $s \geq 0$:

$$\min \quad x_1 + x_2 \quad s.t. \quad 2x_1 + x_2 + s = 4, \quad x_1, x_2, s \geq 0$$

2 Simplex Algorithm

Different kinds of problems may lead to different kinds of solutions



2.1 Problem Formulation

Linear programs are often solved in **equality form**

$$\begin{aligned} \min_x \quad & c^T x \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0 \\ & x, c \in \mathbb{R}^{2n+m}, \\ & A \in \mathbb{R}^{2n+m} \\ & b \in \mathbb{R}^m \end{aligned}$$

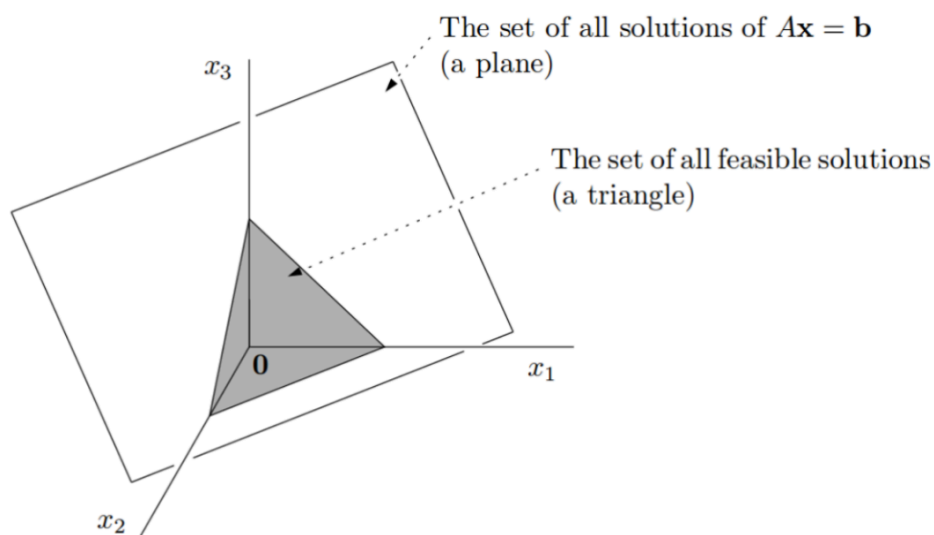


Figure 1: Simplex feasible set

2.2 Info abt the simplex algorithm

- Guaranteed to solve any feasible and bounded linear program
- Works on the equality form
- Assumes that rows of A are linearly independent and $m \leq n'$ ($n' \leq 2n + m$)
- The feasible set of a linear program forms a polytope. (faces of $n - 1$ dimension)
- The simplex algorithm moves between vertices of the polytope until it finds an **optimal vertex**

Theorem (Fundamental Theorem of Linear Programming)

Given:

$$\min \{c^T x \mid x \in P\} \text{ where } P = \{x \in \mathbb{R}^n \mid Ax \leq b\}$$

If P is a bounded polyhedron and not empty and x^* is an optimal solution to the problem, then:

- x^* is an extreme point (vertex) of P , or
- x^* lies on a face $F \subset P$ of optimal solution



Figure 2: Fundamental Theorem of Linear Programming

We can find these vertices using a system of linear equations.

$$Ax = A_B x_B = b \quad \Leftrightarrow x_B = (A_B)^{-1} b$$

for each vertex. Though this is not really done in practice (very costly).

In practice for this is done by LU-Decomposition [LINK] with

$$\mathcal{O}(3m)$$

Where m is the number of constraints

2.3 The Simplex Algorithm

- every vertex has an associated partition (B, N)
- not every partition corresponds to a vertex. A_B might be not invertible or the point x_B might not be ≥ 0 .
- identifying partitions that correspond to vertices corresponds to solving an LP problem as well!

The algorithm consists of two phases

- 1) **Initialization Phase:** finding a feasible starting vertex
- 2) **Optimization Phase:** finding the optimal vertex

2.3.1 The problem setup

The LP

$$\min_x c^T x$$

$$s.t \quad Ax = b \\ x \geq 0$$

$$x, c \in \mathbb{R}^{2n+m},$$

$$A \in \mathbb{R}^{2n+m}$$

$$b \in \mathbb{R}^m$$

Since $n > m$, A vertex is obtained by: $x_N = 0$

- Picking $n - m = 2$ variables to set to zero (the non-basis N)
- Solving for the remaining $m = 2$ variables (the basis B)

$$Ax = A_B x_B + A_N x_N = b$$

Since $x_N = 0$

$$Ax = A_B x_B = b \implies x_B = A_B^{-1} b$$

Lagrangian function:

$$\mathcal{L}(x, \mu \geq 0, \lambda) = c^\top x - \mu^\top x - \lambda^\top (Ax - b)$$

The necessary conditions for optimality (KKT) are also sufficient for linear programs

- feasibility

$$Ax = b, x \geq 0$$

- dual feasibility

$$\mu \geq 0$$

- complementary slackness

$$\mu \cdot x = 0$$

- stationarity

$$A^\top \lambda + \mu = 0$$

3 Duality

Linear programs have a simple dual form:

$$\begin{array}{ll} \min_x c^\top x & \max_\lambda b^\top \lambda \\ \text{s.t. } Ax = b, x \geq 0 & \text{s.t. } A^\top \lambda \leq c \end{array}$$

Let solution to the primal be p^* and solution to the dual be d^* . Then, $p^* = d^*$ guarantees to be the unique optimal value because the duality gap is zero.

This is due to:

$$\begin{array}{ll} \min_x \max_\lambda c^\top x + \lambda^\top (Ax - b) \\ \text{s.t. } Ax = b \end{array}$$

For the case of $Ax \geq b$ then we search for $\lambda \leq 0$

Theorem (Strong Duality Theorem)

Given:

$$(P) \quad \min\{c^\top x \mid Ax = b, x \geq 0\}$$

$$(D) \quad \max\{b^\top \lambda \mid A^\top \lambda \leq c\}.$$

Exactly one of the following occurs:

- 1) Both (P) and (D) are feasible, and $p^* = d^*$.
- 2) (P) is unbounded below and (D) is infeasible.
- 3) (P) is infeasible and (D) is unbounded above.
- 4) (P) has feasible solution, then let an optimal be: $x^* = [x_1^*, \dots, x_n^*]$. Then, (D) is feasible and has an optimal solution λ^* such that $\lambda^* = [\lambda_1^*, \dots, \lambda_m^*]$.

Then

$$p^* = c^\top x^* = b^\top \lambda^* = d^*$$

4 Primal-Dual Hybrid Gradient (PDHG) Method for LP

4.1 First-order methods for LP

For the standard form of the linear program

$$\begin{aligned} \min_x \quad & c^\top x \\ \text{s.t.} \quad & Ax = b, x \geq 0 \end{aligned}$$

we can write the Lagrangian as

$$\mathcal{L}(x, \mu \geq 0) = c^\top x + \mu^\top (Ax - b)$$

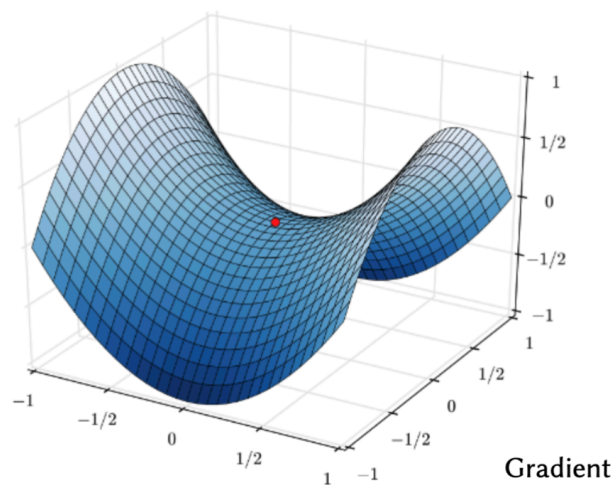


Figure 4: Lagrangian function for LP

Then we can do gradient ascent on the dual variables μ and gradient descent on the primal variables x :

$$\begin{aligned} x^{k+1} &= \text{Projection}_{x \geq 0}(x^k - \alpha_k(c + A^\top y^k)) = (x^k - \alpha(c - A^\top y^k))^+ \\ y^{k+1} &= y^k + \alpha_k(b - Ax^k) \end{aligned}$$

Gradient Descent Ascent may not converge to the saddle point.

4.2 Example:

$$\max_x \min_y xy$$

we have closed form of each iteration:

$$\begin{bmatrix} x_k \\ y_k \end{bmatrix} = \begin{bmatrix} 1 & \alpha \\ -\alpha & 1 \end{bmatrix}^k \cdot \begin{bmatrix} x^0 \\ y^0 \end{bmatrix}$$

The determinant of $\begin{bmatrix} 1 & \alpha \\ -\alpha & 1 \end{bmatrix}$ is $1 + \alpha^2 > 1$ no matter how small α is, which means this approach never converges.

We can fix this issue by updating y^{k+1}

For the original LP problem, usually we use the following metrics for termination:

- Primal feasibility: $\|Ax - b\| \leq \varepsilon$
- Dual feasibility: $\|c - A^\top \lambda\| \leq \varepsilon$
- Duality gap: $|c^\top x - b^\top \lambda| \leq \varepsilon$

Use `method = "highs"` from `scipy.optimize.linprog`